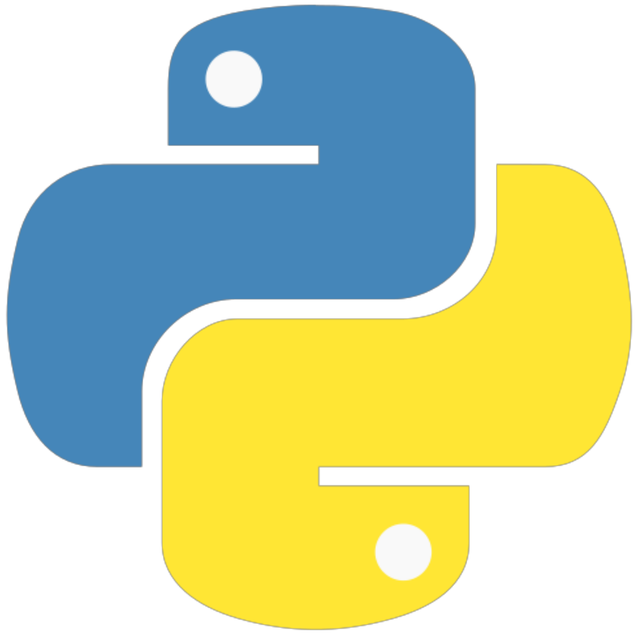
UNIVERSITÄT OSNABRÜCK

# Programmieren für PsychologInnen

## Verschiedenes



Allgemeine Psychologie I
B.Sc. Psychologie / M.Sc. Psychologie - Seminar Programmieren
18.07.2025

Merle Sagehorn, M.Sc.

# Today

- Exkurs: Visual Scripting (in Unreal Engine)
- Offene Fragen & restliche Aufgaben

# Seminar Teilnahme

- Abgabe von Aufgaben bis Ende des Semesters möglich
- Erledigung von mind. 80% der Aufgaben
  - 7 von 9 Aufgaben
- Teilnahme wird auf ToR bestätigt
- Auf Wunsch Teilnahmezertifikat
  - per Mail bei mir melden

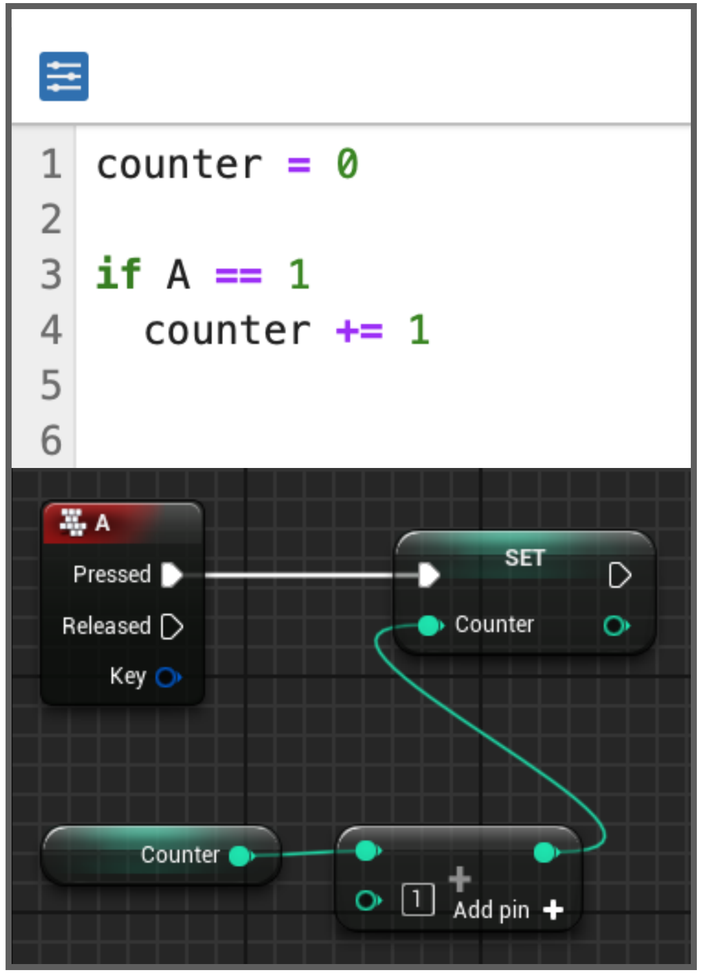
# Visual Scripting

- = Methode zur Erstellung von Spielmechaniken und Logik in Spielen, die auf einer **visuellen**, **knotenbasierten** Darstellung basiert, anstatt auf herkömmlichem Code

- **Unity Visual Scripting:** Ein beliebtes Tool für die Spieleentwicklung mit Unity.
- **Unreal Engine Blueprints:** Unreal Engine's visuelles Skriptsystem.

CODE

NODES

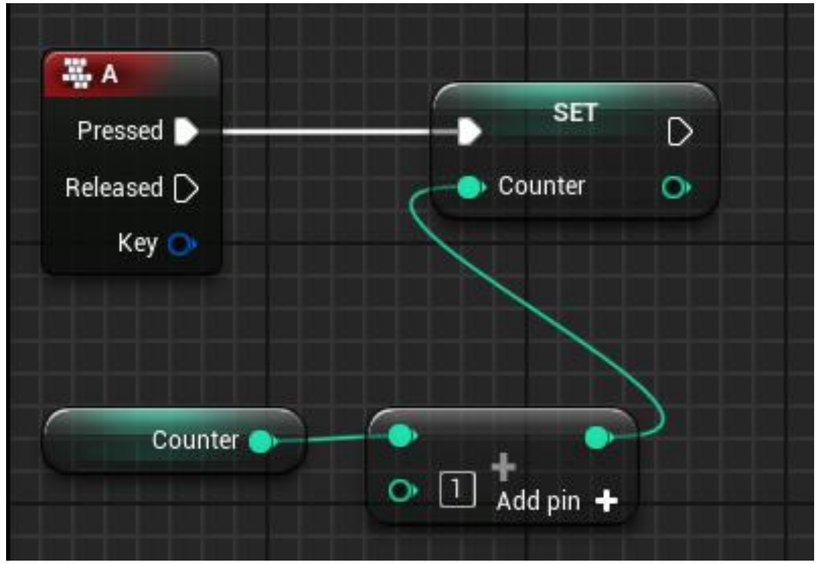


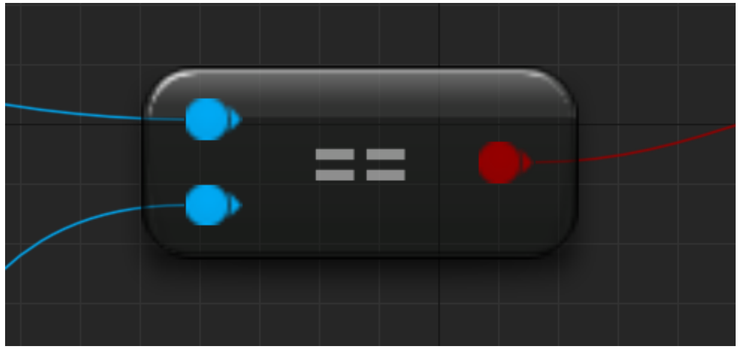
# Unreal Engine

- Grafik-Engine zur Spielentwicklung für Anwendung auf verschiedenen Plattformen
- → PC, Konsolen & Mobilgeräte, VR & AR
- stark objektorientiert und komponentenbasiert
- **visuelles, datenflussorientiertes Scripting**
- Editor-zentriert: Level Editor, Material Editor, Animation Editor, Blueprint Editor
- Echtzeit-Feedback, Kompilierung

# Visual Scripting – Basics – Unreal Engine

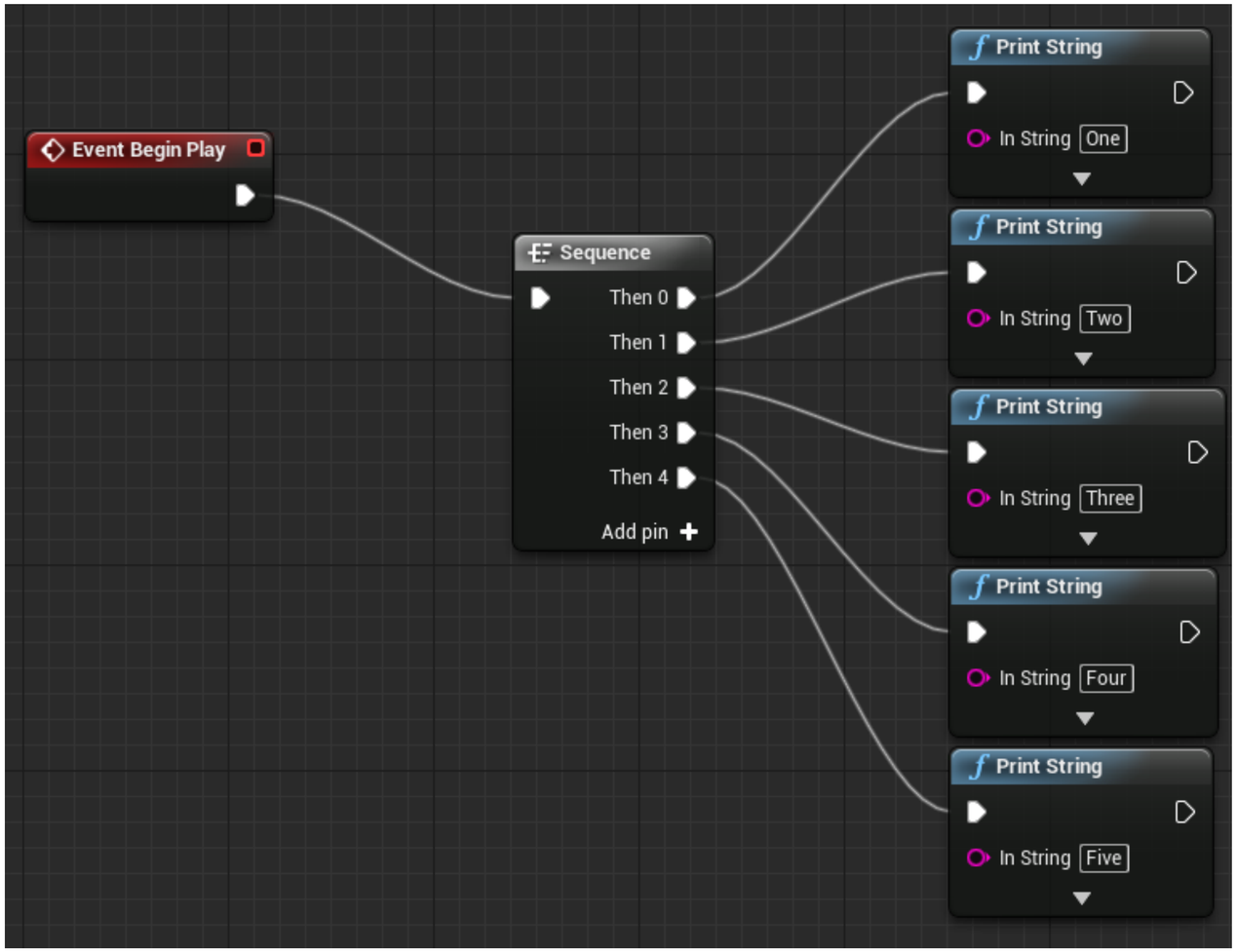
- **Mathematische Operatoren**
- **Relative Operatoren**
- Sequence
- For-Loop
- If-Statement (Branch)
- Debugging



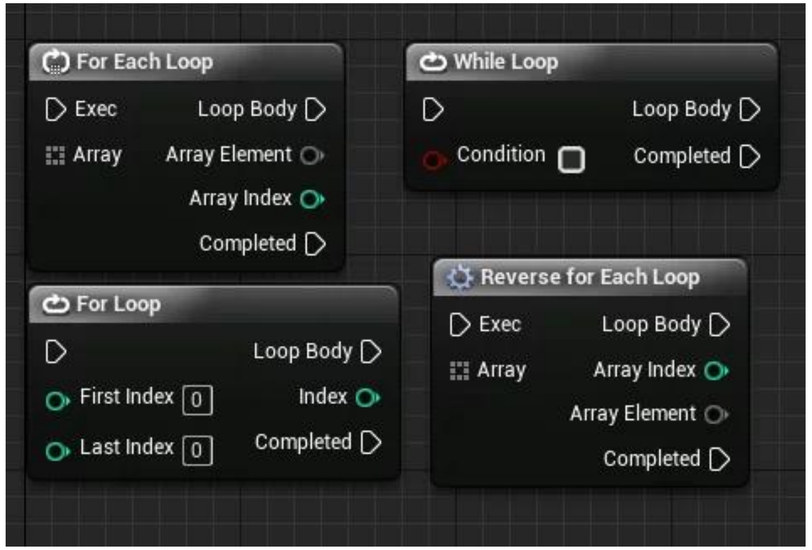
# Visual Scripting - Basics

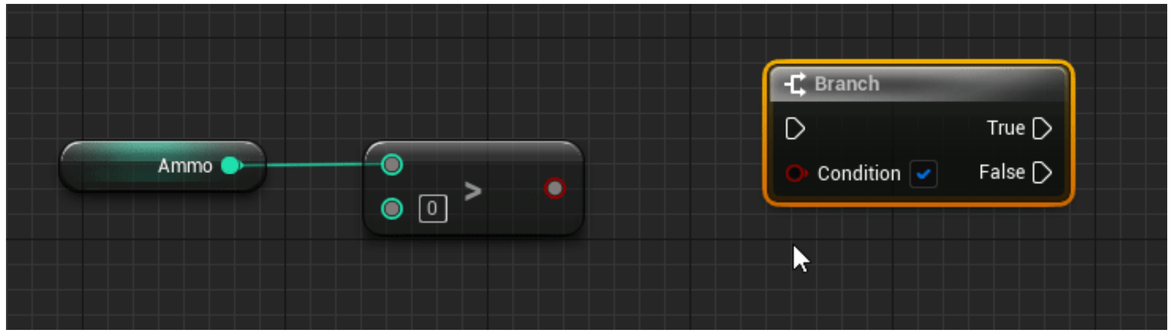
- Mathematische Operatoren
- Relative Operatoren
- **Sequence**
- For-Loop
- If-Statement (Branch)
- **Print**



# Visual Scripting - Basics

- Mathematische Operatoren
- Relative Operatoren
- Sequence
- **For-Loop**
- **If-Statement (Branch)**
- Print



- Animation Avatare
- Buttons
- Türen

File Edit Asset View Debug Window Tools Help

BP_Ada* | BP_Hana

Parent class: Actor

Compile | Diff | Find | Hide Unrelated | Class Settings | Class Defaults | Simulation | No debug object selected

## Components

Add | Search

- BP_Ada (Self)
  - Root
    - Body
      - Feet
      - Legs
      - Torso
      - Face
        - Mustache

## My Blueprint

Add | Search

**GRAPHS**
- EventGraph

**FUNCTIONS** (19 OVERRIDABLE)
- ConstructionScript
- EnableMasterPose
- ARKitFaceSetup
- LiveRetargetSetup
- LiveLinkSetup
- HairLODSetup
- SetUpdateAnimationInEditor

**MACROS**

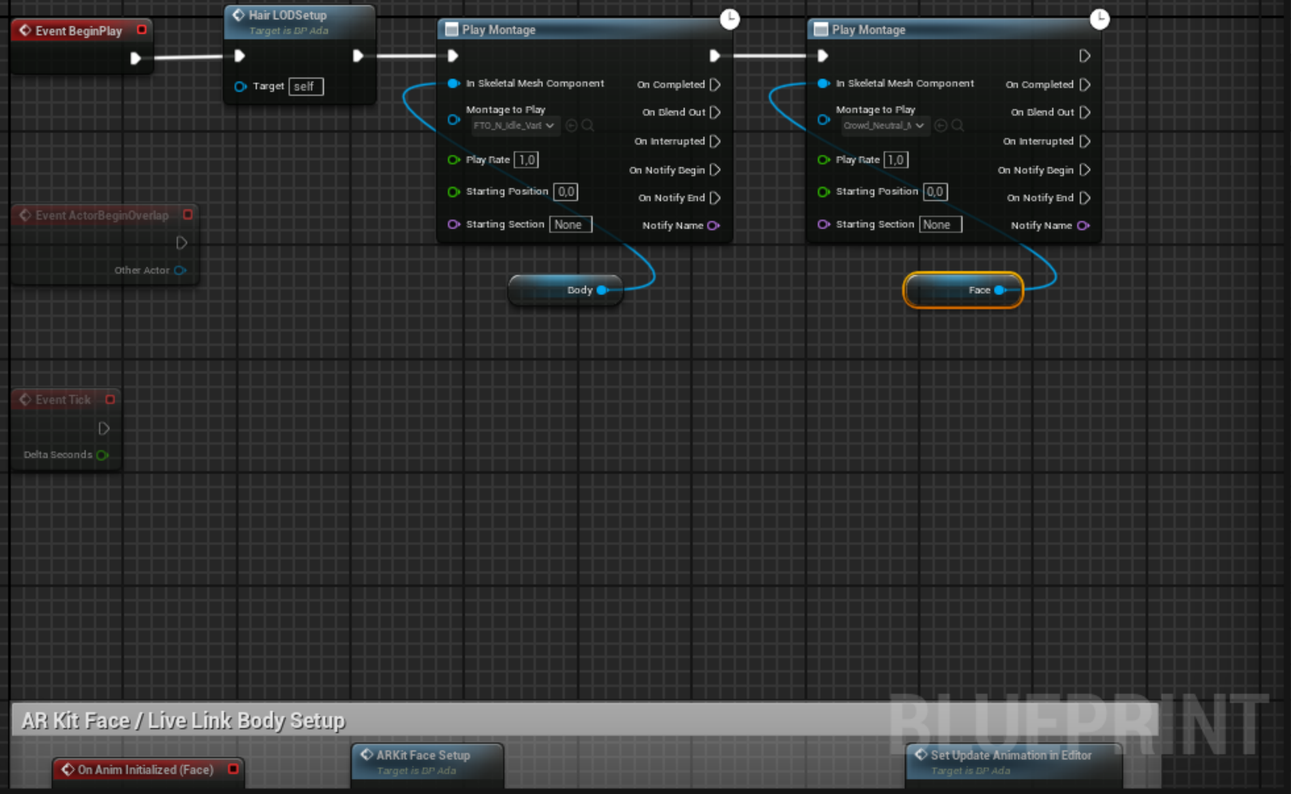
**VARIABLES**
- Components
- Live Link
- Live Retarget

**EVENT DISPATCHERS**

Viewport | Construction Scr... | Event Graph

BP_Ada > Event Graph

Zoom -3




Compiler Results | Find Results

CLEAR

## Details

Search

**Variable**

| | |
|---|---|
| Variable Name | Face |
| Variable Type | Skeletal Mesh |
| Expose to Cinematics | |
| Category | Default |

Advanced

**Default Value**

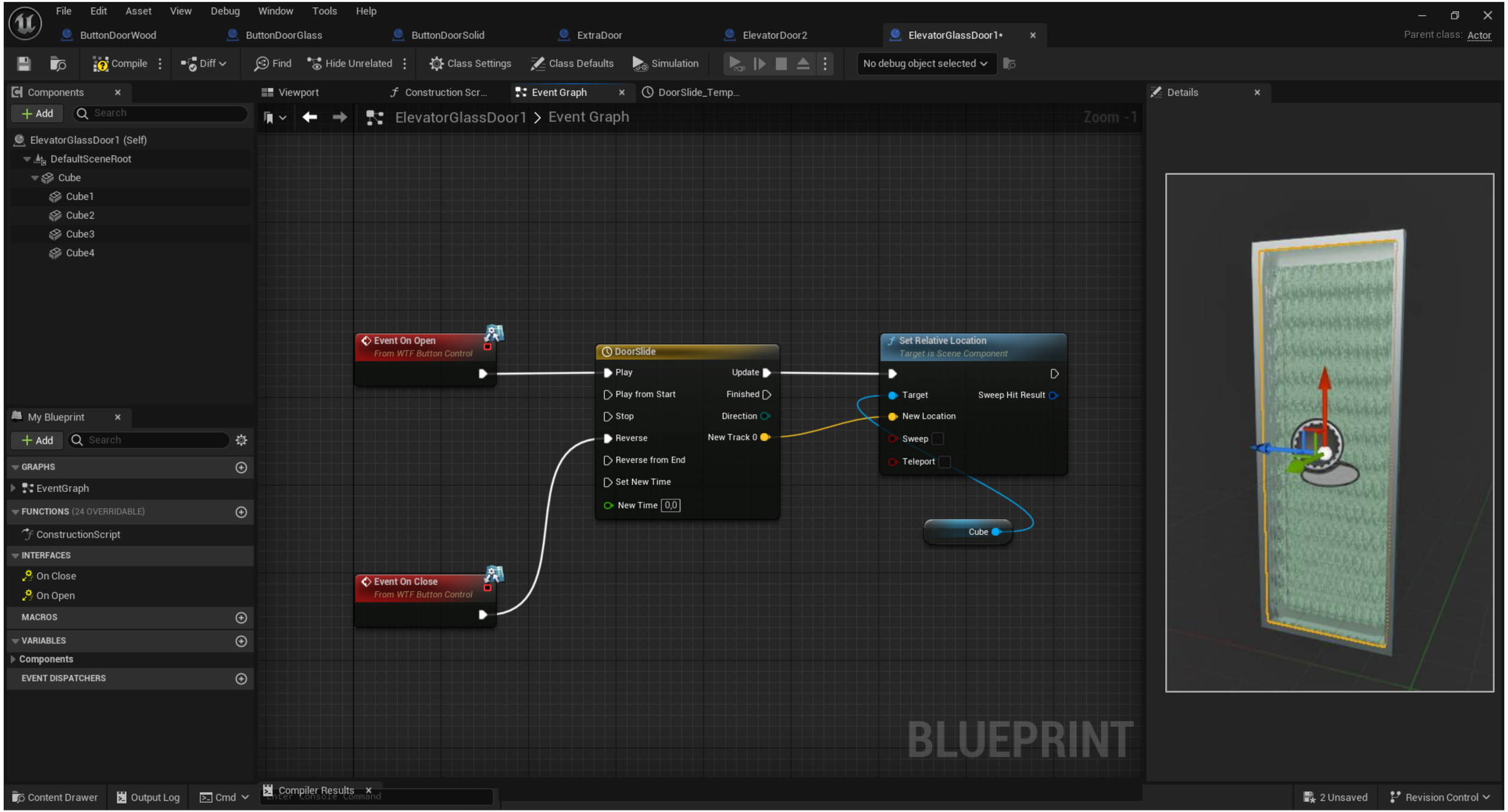
| | |
|---|---|
| Face | None |

**Events**

- On Constraint Broken
- On Plastic Deformation
- On Anim Initialized
- On Component Hit
- On Component Begin Overlap
- On Component End Overlap
- On Component Wake
- On Component Sleep
- On Component Physics State Chang
- On Begin Cursor Over
- On End Cursor Over
- On Clicked
- On Released
- On Input Touch Begin
- On Input Touch End
- On Input Touch Enter
- On Input Touch Leave
- Physics Volume Changed
- On Component Activated
- On Component Deactivated

Content Drawer | Output Log | Cmd | Enter Console Command

1 Unsaved | Revision Control


File
Edit
Asset
View
Debug
Window
Tools
Help
ButtonDoorWood
ButtonDoorGlass
ButtonDoorSolid
ExtraDoor
ElevatorDoor2
ElevatorGlassDoor1*
Parent class: Actor
Compile
Diff
Find
Hide Unrelated
Class Settings
Class Defaults
Simulation
No debug object selected
Components
Add
Search
ElevatorGlassDoor1 (Self)
DefaultSceneRoot
Cube
Cube1
Cube2
Cube3
Cube4
My Blueprint
GRAPHS
EventGraph
FUNCTIONS (24 OVERRIDABLE)
ConstructionScript
INTERFACES
On Close
On Open
MACROS
VARIABLES
Components
EVENT DISPATCHERS
Viewport
Construction Scr...
Event Graph
DoorSlide_Temp...
ElevatorGlassDoor1 > Event Graph
Zoom -1
Event On Open
From WTF Button Control
DoorSlide
Play
Play from Start
Stop
Reverse
Reverse from End
Set New Time
New Time 0,0
Update
Finished
Direction
New Track 0
Set Relative Location
Target is Scene Component
Target
New Location
Sweep
Teleport
Sweep Hit Result
Cube
Event On Close
From WTF Button Control
BLUEPRINT
Details
Content Drawer
Output Log
Cmd
Compiler Results
Enter Console Command
2 Unsaved
Revision Control

File Edit Asset View Debug Window Tools Help

ButtonDoorWood | ExtraDoor | LoadLevelCollider | UnloadLevelCollider | ButtonAnswers* | ButtonDoorGlass | ButtonDoorSolid | CloseDoubleDoor | CloseSingleDoor

Parent class: Actor

Compile | Diff | Find | Hide Unrelated | Class Settings | Class Defaults | Simulation | No debug object selected

Components | Viewport | Construction Scr... | Get Mat Name Old | VPAnswer Correct | Get Mat Name N... | Give Feedback | Event Graph

ButtonAnswers > Event Graph

Zoom -2

ButtonAnswers (Self)

My Blueprint

VARIABLES

Components

| | |
|---|---|
| IsOn | Boolean |
| GreenButtonStart | Vector |
| RedButtonStart | Vector |
| GreenReady4Answer | Boolean |
| RedReady4Answer | Boolean |
| ActorVariable | Actor |
| Levels | Integer |
| LevelCounter | Integer |
| Level01 | Integer |
| ColliderActive | Boolean |
| Level02 | Integer |
| Level03 | Integer |
| Level04 | Integer |
| Level05 | Integer |
| Level06 | Integer |
| Level07 | Integer |
| Level08 | Integer |
| Level09 | Integer |
| Level10 | Integer |
| Level00 | Integer |
| NewStimuli | Integer |
| NewStimulusRef | Material |
| AnswerGiven | String |
| AnswerCorrect | String |
| NewStimCounter | Integer |

EVENT DISPATCHERS

On Component Begin Overlap (GreenButtonCollider)
Overlapped Component
Other Actor
Other Comp
Other Body Index
From Sweep
Sweep Result

Move Component To
Move | Completed
Stop
Return
Component
Target Relative Location
Target Relative Rotation X 0,0 Y 0,0 Z 0,0
Ease Out
Ease In
Over Time 0,2
Force Shortest Rotation Path

Green Button

Green Button Start

X 0,0 Y 2,0 Z 0,0 Add pin

Branch
True | Condition | False

Green Ready 4Answer

SET
Red Ready 4Answer

SET
Answer Given 1

Set Material
Target is Primitive Component
Target
Element Index 0
Material
GreenGlow_Inst

Green Button

SET
Green Ready 4Answ

Rendersorung eher über externen Input

On Component End Overlap (GreenButtonCollider)
Overlapped Component
Other Actor
Other Comp
Other Body Index

Move Component To
Move | Completed
Stop
Return
Component
Target Relative Location
Target Relative Rotation X 0,0 Y 0,0 Z 0,0
Ease Out
Ease In
Over Time 0,3
Force Shortest Rotation Path

Green Button

Green Button Start

Set Material
Target is Primitive Component
Target
Element Index 0
Material
GreenUnglow_Ir

Green Button

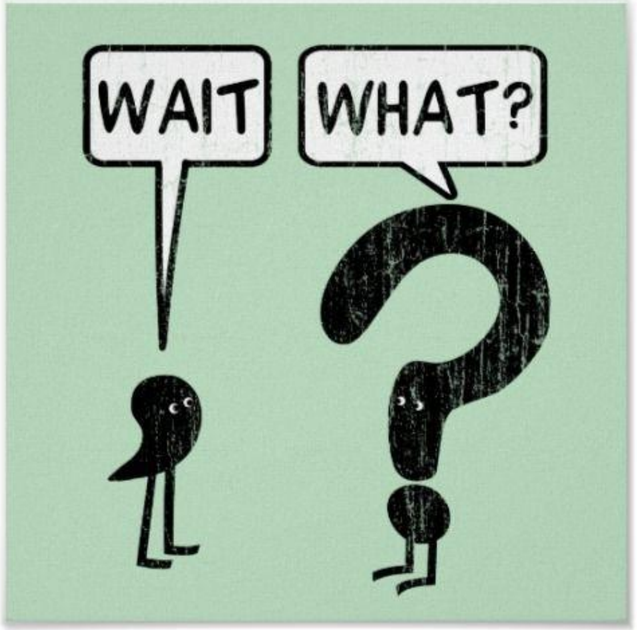
BLUEPRINT

Content Drawer | Output Log | Cmd | Compiler Results

1 Unsaved | Revision Control

# Your turn!

- Offene Fragen? ✓
- Fehlende Aufgaben fertig stellen ✓
- Evaluation des Seminars ☺ ✓



# Evaluation